

MedAgent-Flow: Agentic Architecture with MedGemma

Brief Description: The project consists of developing an Agentic AI ecosystem focused on medical training, utilizing Google's MedGemma foundation model. The system acts as a proactive tutor that plans and executes clinical scenarios based on the Appoint-Ready concept, managing the full flow of a digital consultation.

The group shall work on the following areas:

- Agent Orchestration: Create an architecture where different agents (Diagnosis, Tutor, and Narrator) cooperate to react to the student's decisions.
- Clinical Reasoning with LLMs: Integrate MedGemma to process natural language and ensure that the "virtual patient's" responses are clinically plausible.
- Gamification Mechanics: Implement a scoring and progression system that evaluates not only technical accuracy but also efficiency and empathy in medical dialogue.

Methodology and Data:

- AI and Deep Learning: Implementation of an agent architecture that utilizes MedGemma via API or orchestration frameworks (such as LangChain or AutoGen) for natural language processing and clinical decision-making.
- Data: Large-scale generation of synthetic data and clinical cases using MedGemma itself, ensuring the project is feasible without the need for real hospital data, respecting ethical privacy principles.
- Sensorization and Environment: The "environment" will be the interaction flow on the platform (telemetry logs), where each student action is treated as a sensory signal that feeds the agent's feedback loop.

Expected Results for the UC:

- Functional Prototype: An application or interface where a student can perform a virtual consultation guided by intelligent agents.
- Gamification Engine: Implementation of performance metrics (accuracy, empathy, time) that influence agent behavior and game progression.
- Technical Report: Documentation on the integration of Foundation Models in medical education systems and an analysis of the accessibility and inclusivity of the proposed solution.